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Vision-skeleton dual-modality framework for generalizable assessment of Parkinson's disease gait

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ABSTRACT

Gait abnormalities in Parkinson's disease (PD) can reflect the extent of dysfunction, and making their assessment crucial for the diagnosis and treatment of PD. Current video-based methods of PD gait assessment are limited to only focusing on skeleton motion information and are confined to evaluations from a single perspective. To overcome these limitations, we propose a novel vision-skeleton dual-modality framework, which integrates keypoints vision features with skeleton motion information to enable a more accurate and comprehensive assessment of PD gait. We firstly introduce the Keypoints Vision Transformer, a novel architecture designed to extract vision features of human keypoints. This model encompasses both the spatial locations and connectivity relationships of human keypoints. Subsequently, through the proposed temporal fusion encoder, we integrate the extracted skeleton motion with keypoints vision features to enhance the extraction of temporal motion features. In a video dataset of 241 PD participants recorded from the front, our proposed framework achieves an assessment accuracy of 78.05%, which demonstrates superior performance compared to other methods. To enhance the interpretability of our method, we also conduct a feature visualization analysis of the proposed dual-modality framework, which reveal the mechanisms of different body parts and dual-modality branch in PD gait assessment. Additionally, when applied to another video dataset recorded from a more general perspective, our method still achieves a commendable accuracy of 73.07%. This achievement demonstrates the robust generalization capability of the proposed model in PD gait assessment from cross-view, which offers a novel approach for realizing unrestricted PD gait assessment in home monitoring. The latest version of the code is available at https://github.com/FJNU-LWP/PD-gait-VSDF.

1. Introduction

Gait assessment plays a crucial role in evaluating the severity of Parkinson's disease (PD) in patients (Fox et al., 2018; Sveinbjornsdottir, 2016). By meticulously observing and analyzing the gait performance, physicians can derive deep insights into PD's progression, leading to more precise diagnoses and treatment plans (Mirelman et al., 2019). Currently in clinical practice, the widely accepted assessment of PD gait is based on the contents of Part III of the MDS-UPDRS (Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale) (Evers et al., 2019; Goetz et al., 2023). During the assessment of PD gait, it is imperative that patients comply with specified testing protocols in the MDS-UPDRS to accurately capture the characteristics of their gait. Trained evaluators are required to meticulously observe critical gait indicators and conduct a thorough assessment of gait performance (Goetz et al., 2008; Tilley et al., 2014). However, this method is time-intensive and demands considerable medical resources. Despite the evaluators' professional training and expertise, subjective variations can influence the scoring, introducing a significant element of subjectivity. Consequently, there is an urgent need for an objective and precise approach to assessing gait in PD patients.

Utilizing computer-assisted methods for the assessment of PD gait is

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1361-8415/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

a hot topic in the research of PD motor symptom evaluation. In recent years, numerous studies have explored various automated technologies to quantify gait motion of PD patients based on wearable sensors (Hynes et al., 2021; May et al., 2023). Some sensor-based methods of gait assessment have demonstrated preliminary capabilities in identifying diverse gait characteristics by collecting data from patients with PD. These methods enable the monitoring of critical gait motion indicators, such as gait speed, stride duration, and coordination (Lu et al., 2020b; Moreau et al., 2023). Such advancements furnish a robust framework for clinical decision-making and the treatment of PD. Furthermore, the integration of sensor technology with sophisticated data processing techniques suggests a potent approach for gaining insights into and assessing motor function impairments in PD (Espay et al., 2016; Zhong et al., 2021). However, these methods rely on sensors that directly contact the patient's body, inevitably affecting the PD patient's natural movement, which has hindered their widespread adoption in clinical practice.

Recent advances in computer vision have enabled non-contact video-based methods for PD gait assessment that overcome limitations of sensor-based approaches. Video technology offers a contactless, scalable, and non-invasive approach for the assessment of PD, hence its widespread application in the identification, quantification, and tracking of gait characteristics in patients with PD (Archila et al., 2022; Chandrabhatla et al., 2022). Guo presented a novel approach to PD gait assessment through a two-stream spatial-temporal attention graph convolutional network (2s-ST-AGCN) (Guo et al., 2022b). Mark Endo introduced GaitForeMer, a Transformer-based model that employs human motion forecasting as a self-supervised pre-training task for estimating the severity of gait impairments in PD (Endo et al., 2022). Mohsen Gholami proposed a weakly-supervised learning approach for automatic labeling of PD gait videos, utilizing domain-specific knowledge from neurologists without the need for a priori MDS-UPDRS ratings (Gholami et al., 2023). These studies utilized video technology, supplemented by deep learning and human pose estimation algorithms, to accurately quantify human motion. They have demonstrated that contactless video analysis technology can effectively and rapidly assess the gait of patients with PD. However, most of the current video-based methods only rely on skeleton information obtained by human pose estimation in videos, neglecting the vision features of PD gait. Moreover, they are only applied to PD gait assessments from a single recording perspective, demonstrating limited generalizability.

In light of the aforementioned issues, we initially consider leveraging the vision information from PD gait video. In multimodal action recognition based on skeletons, both skeleton motion information and RGB image data are typically utilized (Yu et al., 2023; Zhu et al., 2022). To the best of our knowledge, most existing video-based PD gait assessment methods rely on mid-level features such as optical flow or pose estimations (Archila et al., 2022; Guayacan et al., 2022), which may discard certain visual information from the original RGB images during the extraction process. However, directly using the entire RGB information from videos introduces unnecessary background details and significantly increases unnecessary computation (Eguchi et al., 2023). In fact, evaluators primarily focus on the status of individual body parts (Mirelman et al., 2019; Muller et al., 1997). Consequently, we propose a novel transformer-based model to extract vision features of human keypoints. Unlike patches used in conventional transformer models, our patches of human keypoints incorporate positional information. Consequently, we embed the position information of each keypoint directly within the corresponding patch. Furthermore, considering the connectivity between human keypoints, we introduce the adjacency embeddings in the model to incorporate their connection information. Through the proposed model, we can obtain PD gait features from a more detailed perspective. However, the assessment of PD gait is a comprehensive process (Lee et al., 2016; Vervoort et al., 2013). Although this proposed model can extract gait characteristics from a novel way, further consideration is warranted on how to effectively utilize more macroscopic skeleton motion features.

Based on this, we consider extracting the features of local human keypoints and subsequently fusing skeleton motion information. We first fuse the two types of features within each temporal unit, and then extract the temporal dynamics of gait through the proposed encoder. Through the above approach, we can assess PD gait from a more comprehensive perspective. Fig. 1 showcases the process of our generalizable gait assessment method. Initially, videos of the PD gait assessment were recorded by a standard camera from a frontal perspective. Concurrently, three raters evaluated the participants' gait performance, and the majority vote from these evaluations served as the ground truth for the corresponding video. Following this, our proposed vision-skeleton dual-modality framework was trained using a supervised learning approach. Ultimately, the trained model was applied to gait videos recorded from other perspectives to validate its generalizability in cross-view PD gait assessment. Overall, we propose a model based on the keypoints vision and skeleton motion. Compared to other methods, our approach demonstrates superior performance in PD gait assessment and exhibits excellent generalization capability in cross-view assessments. The key contributions of our method are as follows:

1) We propose a dual-modality framework for video-based PD gait assessment, which integrate the keypoints vision features with skeleton motion information. Compared to other methods, our proposed methods exhibit superior performance and demonstrate exceptional generalization across more general perspectives.
2) We utilize the vision information to the PD gait assessment for the first time. The Keypoints Vision Transformer (KVT) was proposed to extract keypoints vision features from PD gait videos. It incorporates positional information of the keypoints and introduces their connectivity information through adjacency embeddings (AE).
3) We integrate keypoints vision features and skeleton motion information through a temporal fusion encoder, which additionally extracts the temporal dynamics of gait.
4) We demonstrate the interpretability of our proposed framework through the features visualization results of its two branches, which also reveal the mechanisms of different body parts in the assessment of PD gait.

2. Related work

With the continuous advancement of computer vision technologies, human pose estimation algorithms have achieved significant development, allowing current technologies to extract skeleton motion information from video data with high precision (Artacho et al., 2020; Cao et al., 2017). This extracted information provides a technical foundation for the in-depth analysis of gait motions in videos (Deng et al., 2024; Sabo et al., 2020). Particularly in the domain of PD diagnosis and assessment, gait analysis through video capture has emerged as a novel, non-invasive assessment method. Currently, most video-based PD gait assessment methods rely on advanced human pose estimation algorithms. These algorithms can accurately extract skeleton keypoints from complex video data, and subsequent analysis of this skeleton motion information enables the effective evaluation of PD gait characteristics (Sidoroff et al., 2021; Zhao et al., 2024). Michael H. Li is the pioneer in applying deep learning algorithms for the assessment of PD based on visual cues (Li et al., 2018). Utilizing random forest algorithms, their approach has predicted both the overall MDS-UPDRS scores and the scores of Part III of MDS-UPDRS based on motor features. Some gait assessment methodologies have demonstrated potential applicability to tasks related to bradykinesia in hand movements. Mandy Lu introduced a Sequential Focused Neural Network, utilizing human joint information as input to the model, which achieved satisfactory results in predicting the MDS-UPDRS scores for both PD gait and hand bradykinesia (Lu et al., 2021). Despite significant progress in the assessment research of PD gait through video, the lack of a unified standardized assessment protocol

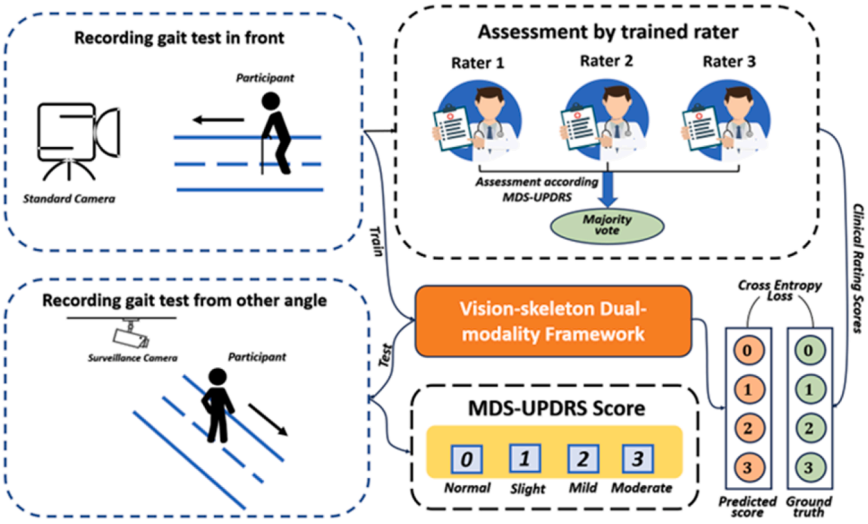


Fig 1. The process overview of generalizable assessment of PD gait in our study.

and broadly recognized validation results limits its application in clinical practice.

With the evolution of human pose estimation algorithms and Graph Neural Networks (GNNs), researchers have increasingly utilized human keypoints as inputs for GNNs to intricately analyze the relational dynamics between movements (Duan et al., 2022; Lee et al., 2023). This approach leverages the structural and sequential information embedded in human movements, enabling a more nuanced understanding of the interaction among various body parts. To address limitations in representation and generalization associated with current methods, Haoyu Tian introduced a spatio-temporal attention enhanced gait-structural graph convolutional network (AGS-GCN), which constructed a gait skeleton graph based on clinical priors and sensor reliability, employing a novel partition strategy and a soft attention mechanism for fine-grained feature extraction and improved performance on diverse datasets (Tian et al., 2022). Evaluating PD motor dysfunction using the MDS-UPDRS is a labor-intensive, costly process, prone to inter-rater variability. To solve this problem, Mohsen Gholami introduced a weakly-supervised approach that classifies gait in PD patient videos without relying on prior MDS-UPDRS ratings, instead utilizing domain-specific insights from neurologists and a novel weakly-supervised 3D pose estimation technique to achieve a marked accuracy (Gholami et al., 2023).

The aforementioned video-based methods for assessing PD gait primarily rely on skeleton data, which may overlook other crucial gait information present in the videos. Consequently, some studies have begun to explore the integration of skeleton information with other gait data from the videos. Qingyi Zeng introduced an innovative automated video-based method for PD gait assessment. This method utilized a novel skeleton-silhouette fusion convolutional network, enabling the quantification of gait features in natural and home-based environments (Zeng et al., 2023). However, despite the utilization of additional features in the video, silhouette characteristics only offer coarse outlines of gait, and still fail to capture motion information of some critical keypoints. Additionally, some studies have combined video-based methods with other approaches for a comprehensive evaluation of outcomes. They employed special sensors to capture 3D gait information, which was then integrated with 2D gait data captured by cameras, demonstrating the correlation between these methodologies (Dong et al., 2023; Jiang et al., 2024; Sabo et al., 2022). These methods indicate that vision-based systems can be utilized as effective instruments for the continuous monitoring of PD patients in residential care environments. However, to the best of our knowledge, the information of local keypoints as a crucial assessment metric in PD gait evaluation has not yet been utilized by existing methods. Therefore, in our research, we focus on the vision information of keypoints. By devising a method to fuse it with traditional skeleton motion information, we demonstrated its generalizability across multi-view gait video data.

3. Materials and methods

3.1. Participants and dataset

The PD gait video data utilized in this study were collected from 241 PD patients over the period from 2019 to 2024. All participants were diagnosed according to the United Kingdom Brain Bank Diagnostic Criteria for PD (UKPDSBB), as established by the PD Society Brain Bank (Marsili et al., 2018). Given the human ethics considerations involved in this study, formal approval was obtained from the Ethics Committee of the First Affiliated Hospital of Fujian Medical University. Furthermore, informed consent was signed by each participant. In this study, all participants were assessed using the MDS-UPDRS in an off-medication state and were recorded the gait assessment on video. To protect patient privacy, these video materials will not be made public (Liu et al., 2023). All PD gait assessment recordings were made by a Sony camera, set to a resolution of 1920 × 1280 and a frame rate of 30 frames per second, with the videos ultimately saved in MTS format. Under the supervision of professional evaluators, each patient underwent a specific PD gait assessment recording. During the PD gait assessment, three evaluators positioned themselves in line with the camera to observe and record the patients' performance. After the gait assessment, they independently assigned scores based on the MDS-UPDRS 3.10 gait assessment criteria. All evaluators participating in our study were neurologists with over 15 years of clinical experience, specializing in movement disorders with a particular focus on PD. They had completed the official training program provided by the International Movement Disorders Society for MDS-UPDRS scoring and held certifications demonstrating their competency. For each video, scores were independently provided by these three specialists. To ensure consistency and fairness, the final label for each video was determined based on a majority vote among the three expert ratings. Table 1 presents the distribution of gait video data scores and gender among the 241 participants collected in this study. The labels for all video data—covering the training, validation, and test sets used from Table 1—were generated using this same evaluation process. Consistent with most current research on PD gait assessment (Kaur et al., 2023; Lu et al., 2020a), no instances of a gait score of 4 were observed among the participants in this study. Therefore, our research focuses exclusively on the four scoring categories ranging from 0 to 3.

Table 1
Distribution of original and augmented gait video data from 241 participants in this study. The ground-truth for each video is determined by the majority vote of raters.

MDS-UPDRS Score	0	1	2	3	All
Male/Female	13/17	39/52	55/50	6/8	113/127
Original	30	91	105	15	241
Augmented	90	273	315	45	723

Despite our study being based on a dataset of gait assessment videos of PD with a considerable number of participants, we still employ data augmentation on the collected video data. Gait as compared to other primary motor symptoms of PD, constitutes an action of a larger amplitude (Reich and Savitt, 2019). Consequently, gait videos can still retain their motion information effectively at a relatively lower frame rate. Based on this understanding, to enhance the utilization efficiency of gait video data, we expand the original gait video dataset through interval sampling methods. By resampling the videos, we expand each original 30 FPS video into three new videos at 10 FPS. Consequently, the resulting video data are tripled in size relative to the number of participants. We request the evaluators to assess the augmented videos, and their scoring results remained accurate which confirms the effectiveness of the video augmentation.

3.2. Overall framework

Fig. 2 illustrates the flowchart of our proposed method in this paper. Firstly, vision and skeleton information of keypoints is extracted from gait videos by human pose estimation. For the keypoints vision, we incorporate their coordinates as the input of Keypoints Vision Transformer (KVT). This model is designed to extract keypoints vision features from PD gait video. Subsequently, the model fuses obtained keypoints vision features with skeleton motion features and inputs them into the Temporal Fusion Encoder, which extracts gait and temporal motion features. Ultimately, the model outputs predicted PD gait score results through a multi-layer perceptron head.

3.3. Keypoints vision transformer

We introduce a novel transformer model, focusing on extracting vision features of keypoints in gait videos. Fig. 3 illustrates the structure of the Keypoints Vision Transformer model. The model first transforms images patches of keypoints into token embeddings. Upon retrieval to the tokens corresponding to the keypoints, further augmentation is performed by adding positional information of the keypoints. Specifically, the positional data are mapped to an appropriate dimension via linear projection, then concatenated with keypoint tokens. This process yields keypoint tokens that are augmented with positional information. Subsequently, by incorporating Adjacency Embedding (AE), the model is endowed with the connectivity relations between human body keypoints. Given that the PD gait assessment task in this study is inherently a classification task, a classification token (CLS Token) is appended at the beginning of the token sequence. Finally, these human body keypoint vision tokens are fed into the designed Keypoints Vision Encoder, which is composed of Keypoints Self-Attention mechanisms.

3.3.1. Keypoints patches embedding

Most of current video-based methods for PD gait assessment are based on the skeleton motion (Ma et al., 2024; Tian et al., 2024). They extract gait features from a macroscopic perspective without incorporating any vision information from the video. However, they might ignore the local motion information of keypoints, which is also one of

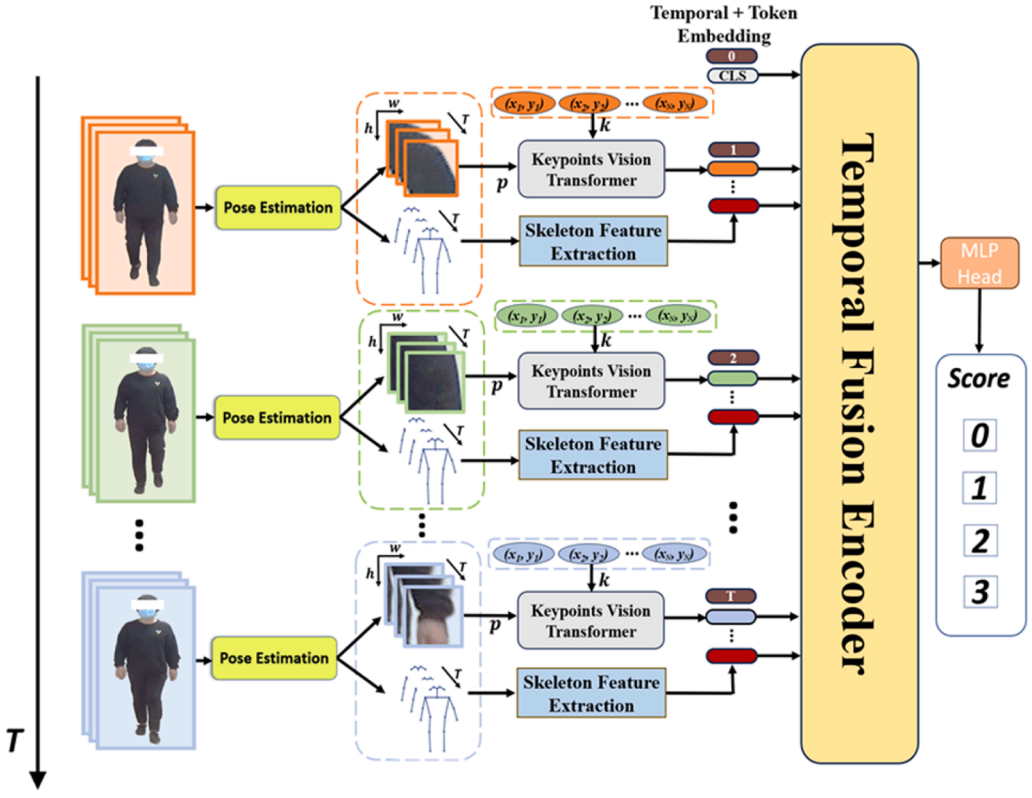


Fig. 2. Our proposed vision-skeleton dual-modality framework operates as follows: Keypoints vision data and skeleton motion data are obtained through pose estimation. The keypoints vision data are input into KVT augmented with coordinates information. After extracting skeleton features, they are combined with the extracted keypoints vision features. Temporal dynamics of the features are captured through the Temporal Fusion Encoder. Ultimately, the final assessment score is outputted by a multilayer perceptron (MLP).

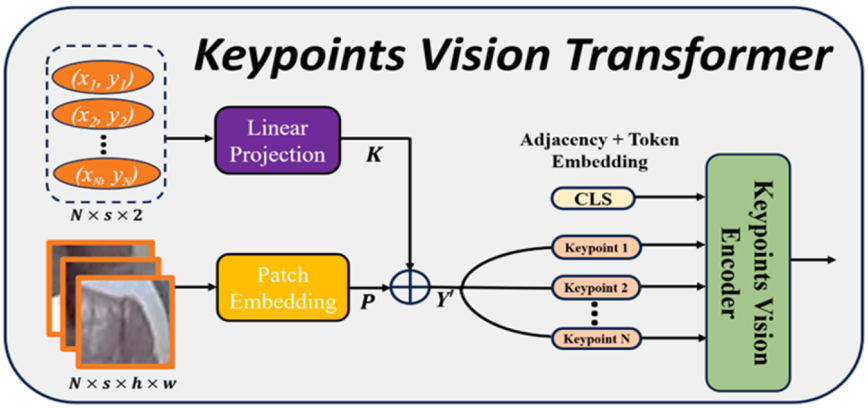


Fig. 3. Keypoints vision transformer:After patch embedding of keypoints vision data, the keypoints coordinates are linearly projected and combined with the results of the patch embedding to generate keypoints tokens. Following concatenation with adjacency embedding and classification token, the combined results are inputted into the Keypoints vision encoder.

the important factors for gait assessment (Janner Zanardi et al., 2021; Konig et al., 2016). Therefore, different from most existing methods, we introduce a novel transformer to extract the vision features of PD gait. Specifically, after obtaining position information of keypoints by human pose estimation, we extract vision features of keypoints. To enable the proposed transformer model to effectively extract keypoints vision features, we first extract vision patches of keypoints from the gait video. These patches are then embedded into tokens for further processing.

We design two approaches to obtain the vision patches of keypoints from a PD gait video $V \in R^{(t\times\ h\ \times w\times c)}$, where t is the number of frames in the input video. A straightforward approach is to get image patches of keypoints from every frame of video, as illustrated in Fig. 4. First, we acquire a sampled video frame Ft from the original video V. Utilizing the positional information obtained from human pose estimation, we subsequently extract local image patches of dimensions $h \times w \times c$ within the regions of keypoints from each Ft, where h and w denote the height and width of patches, c represents the number of RGB channels, which is here fixed at 3. Consequently, we obtain image patches of keypoints $p \in R^{(N\times\ h\ \times w\times c)}$, where N is the number of keypoints. Subsequently, each patch undergoes a linear projection to obtain the Keypoints Tokens (KT). Overall, this method is remarkably straightforward and direct. However, it can only capture static spatial features.

To extract more motion features of keypoints, we extend the patch extraction into the temporal domain, as illustrated in Fig. 5. In this approach, we capture frame sequence patches of size $s \times h \times w \times c$ within the range of each keypoint in Ft, where s represents the number of frames in a short-term sequence. Upon acquiring the local video patches of the keypoints $p \in R^{(N\times s\times\ h\ \times w\times c)}$, we employ 3D convolution to extract features indicative of short-term variations and subsequently generate KT $P \in R^{(T\times N\times C)}$, where T is defined as t/s, C represents the dimensionality of each token. This method can encompass temporal-spatial information over a short duration while also incorporating data from more frames without altering the feature shape.

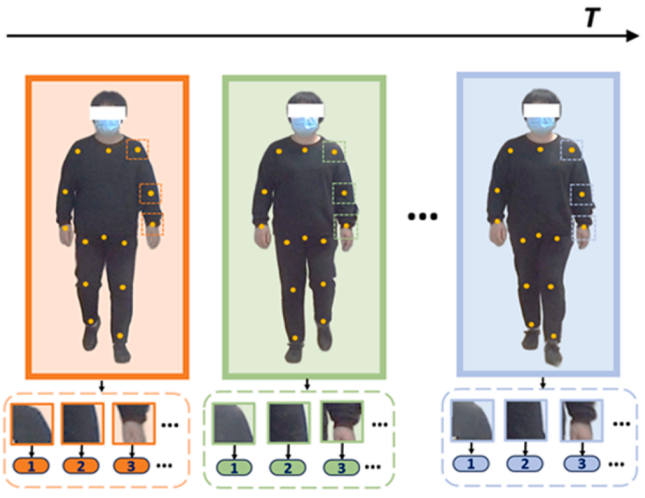


Fig. 4. Image patches embedding of keypoints: We sample N frames and independently embed the 2D frame for each body keypoint.

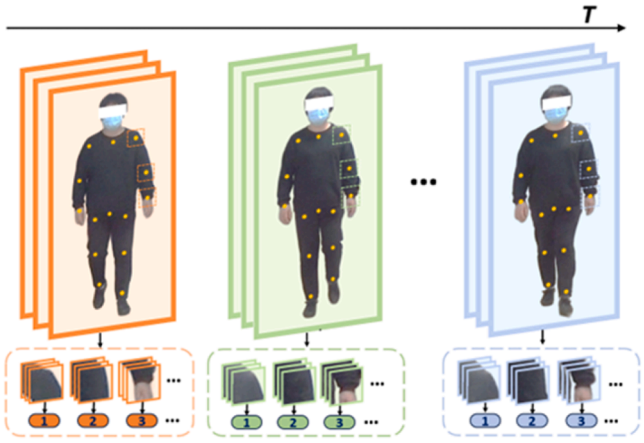


Fig. 5. Frames sequence patches embedding of keypoints: We sample short-duration frame sequences and independently embed frame sequence of each body keypoint.

3.3.2. Positions and connections embedding of keypoints

Given that each body keypoint in the video block possesses specific coordinate information, upon obtaining the KT, we incorporate the positional information of each keypoint into every token of the KT. It is worth noting that, instead of using explicit positional encodings as in standard Vision Transformers (Dosovitskiy et al., 2020), we embed the coordinates of keypoints through a learnable linear projection. This approach simplifies the input pipeline by mapping the raw coordinates directly into the same feature space as the input tokens. Compared to positional encodings, which are typically designed for dense, grid-like data such as image patches, linear projection is more suitable for modeling the sparse and graph-structured nature of human skeletons, offering a more adaptable and structurally aligned representation. For KT $P \in R^{(T\times N\times C)}$, its corresponding 2D skeletal average coordinate data is denoted as $k \in R^{(T\times N\times s\times\ 2)}$. To align the dimensions of these two datasets, we perform a linear projection of the coordinate dimensions of

k. After linear projection, we obtain $K \in R^{(T\times N\times C)}$, which is the transformed feature vector of human body keypoint coordinates. Upon acquiring the linearly projected human body keypoint coordinate data, we concatenate it with P to obtain:

$$Y' = Concat(P, K) \tag{1}$$

Herein, $Y' \in R^{(T\times N\times 2C)}$ represents the KT that incorporates coordinate location information. Positional embeddings are crucial in Transformer architectures for providing spatial context to image patches due to the model's lack of inherent sequential processing (Han et al., 2023). However, our work focuses on the keypoints patches from gait video, which, unlike other images, do not merely possess fixed positional information but also include connectivity information among keypoints. That is, the patches encompass connectivity information. Consequently, we introduce AE based on the adjacency matrix between keypoints to represent connectivity among Keypoint Tokens. The size and numerical values of the adjacency matrix are actually determined by the number of keypoints and their connectivity derived from the adopted pose estimation method (Hassan and Ben Hamza, 2023; Li et al., 2023). In our implementation, we used the skeleton structure consistent with the VIBE framework. From the set of keypoints provided by VIBE, we further selected 14 essential keypoints (as indicated in Figs. 4 and 5) to construct the adjacency matrix. The resulting adjacency matrix A is a binary 14×14 matrix, where each entry $A(i.j) = 1$ indicates a direct anatomical connection between keypoints i and j. To facilitate the integration of the adjacency matrix with the tokens of KT, the adjacency matrix A is transformed through a linear projection followed by an activation function:

$$AE = ReLU(A\cdot W' + q) \tag{2}$$

Following the aforementioned transformations, AE is obtained, where W' denotes the weight matrix, q represents the bias term, and ReLU is the activation function. The application of Adjacency Embedding is akin to positional embedding, taking into account the spatial connections between human body keypoints in the computation of self-attention. This approach enables the model to not only understand the individual features of keypoints but also capture the interactions and relative positional information among them.

Upon acquiring the AE, it is added to Y', which contains the coordinate positional information, resulting in the final KT that incorporates Adjacency Embedding. Moreover, we add a CLS token at the beginning of these Token sequences post-generation of all KT. This token serves as a global representative that aggregates information across all keypoint tokens during the Transformer encoding process. After the self-attention layers, the output of the CLS token is used for the final prediction, as it captures the global spatiotemporal context of the input sequence. In processing sequence data, the CLS token is appended at the start of the input sequence to capture the global information of the entire sequence. Throughout the training process, the CLS token is updated across the model's layers to encompass the summary information of the entire input sequence. Ultimately, the output of the CLS token is utilized for sequence-level classification tasks. This process can be described as follows:

$$Y = [CLS, Y' + AE] \tag{3}$$

Herein, $Y \in R^{(T\times (N+1)\times 2C)}$ represents the KT that includes adjacent embeddings and has been augmented with a CLS token. Upon obtaining an operable Y, it is fed into a specialized encoder designed for keypoint vision features.

3.3.3. Keypoints self-attention

After keypoints vision embedding, we obtain the keypoints vision tokens, which are further used to extract features through proposed encoder. For the input $Y \in R^{(T\times (N+1)\times 2C)}$, it is transformed into queries Q, keys K, and values V based on the fundamental formula of the self-attention mechanism (Dosovitskiy et al., 2020). This transformation is achieved through linear projection:

$$\begin{cases} Q = YW^Q \\ K = YW^k \\ V = YW^V \end{cases} \tag{4}$$

Herein, W^Q, W^k, and W^V represent the weight matrices (Vaswani et al., 2017). Given that the input data in this study consist of information between human body keypoints, we improve upon the self-attention component of the encoder by introducing the Keypoints Self-Attention (KSA). In this keypoints vision encoder, the input keypoints vision Tokens already include the positional coordinates of the corresponding keypoints in video images.

To enhance the encoder's capability to understand and process the spatial relationships between joints within the input data, we introduce AE on top of self-attention mechanism, thereby adequately considering the connectivity among human body keypoints:

$$Keypoints\ SelfAttention(Q, K, V) = SoftMax\left(\frac{QK^T}{\sqrt{d_k}} + AE\right)V \tag{5}$$

Here, the similarity scores between queries and keys are computed by $\frac{QK^T}{\sqrt{d_k}}$, followed by the application of the SoftMax function to obtain normalized attention weights. The term d_k denotes the dimension of the key vectors. Subsequently, these attention weights are multiplied with the values V to generate the final output, which is then forwarded to the next layer of the network. The KSA represents an innovative approach to human body keypoints self-attention, fundamentally utilizing adjacent embeddings to adjust attention weights. Its structure is depicted in Fig. 6. In the context of processing gait videos of PD, KSA incorporates physical connectivity information between human body joints via AE, thereby capturing and understanding the dynamic interactions among human body keypoints with greater precision.

3.4. Skeleton features extraction

After obtaining the vision features of human keypoints, we use a basic ST-GCN to extract the skeleton features from PD gait videos. These features are subsequently embedded into the input of the subsequent encoder. Specifically, upon obtaining the skeleton data of PD gait, we extend graph convolution to the spatio-temporal dimension by connecting the same keypoints across consecutive frames. This approach incorporates not only spatial connections but also integrates temporal dimensions into the analysis (Yan et al., 2018). The function of ST-GCN can be formulated as follows:

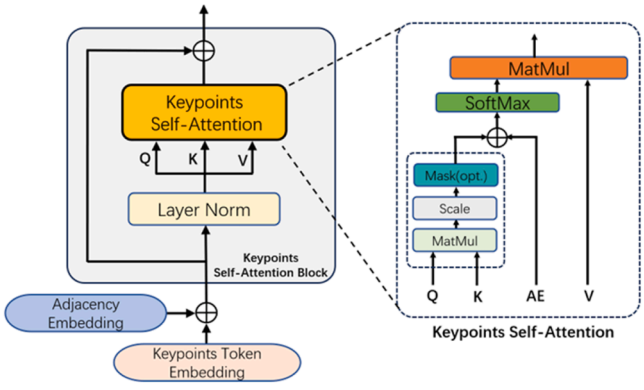


Fig. 6. Keypoints Self-Attention: We incorporate AE into the self-attention. After obtaining the standard dot-product attention, it is combined with AE. This addition allows the computation of attention weights to consider the connectivity information among keypoints.

$$STGCN(f_{in}, A, M) = \sum_{\tau j} \frac{1}{z_{ij\tau}} W(l_{ij}, \tau) \cdot f_{in}^{(t+\tau)}(v_j) \odot M \quad (6)$$

Here, f_{in} and f_{out} respectively denote the input and output feature functions of each keypoint, $z_{ij\tau}$ is a normalization factor. $W(l_{ij}, \tau)$ is the learned weight for the edge type l_{ij} over temporal offset τ. The operation $\odot$ indicates element-wise multiplication. After this, the model incorporates the learning of edge importance by introducing a learnable mask M in each layer of graph convolution (Wu et al., 2019). This mask adjusts the contributions of edges to the feature representations. In this formulation, A denotes the adjacency matrix. In this process, a fully connected layer is utilized to transform the extracted features so that their dimensions match the output of the keypoint vision encoder, thereby forming new tokens.

3.5. Temporal fusion encoder

After embedding the features of keypoints vision and skeleton motion, we consider fusing their respective feature tensors. In our research, both types of features are temporally synchronized to ensure feature alignment. We experiment with several regular fusion methods and find that concatenation produces the most effective results. This fusion way is straightforward, and it allows the complete preservation of both original feature sets. It should be noted that the fusion here is conducted within a short temporal span. To extract the temporal dynamics of the fused gait features throughout the entire video, we introduce the Temporal Fusion Encoder. This encoder is capable of integrating local keypoints vision features and global skeleton motion features at different times throughout the video and extracting their temporal dynamic characteristics.

In this study, we process PD gait video data, where the temporal dimension is a critical factor, as it represents the sequence and dynamic changes between images. Consequently, we also introduce Temporal Embeddings (TE) to enhance the model's capability to process temporal dynamic information in PD gait video. This mechanism encodes the temporal position of each video frame in a manner similar to position encoding. Additionally, the design enables the model to recognize not only the vision information within the videos but also to comprehend the movement of this content over time. We incorporate the TE into the input token sequence through direct addition. By incorporating TE, the encoder becomes capable of comprehensively considering both spatial and temporal information in PD gait videos, thereby enhancing the efficacy of extracting and assessing the characteristic manifestations of PD gait.

To further optimize the model and better integrate the temporal features between frames, a temporal CLS token is added at the beginning of the input token sequence for the temporal fusion encoder. Consequently, the final input to the temporal fusion encoder, denoted as $Input_{TF}$:

$$Input_{TF} = [CLS_{TF}, F' + TE] \quad (7)$$

Here, F' denotes the tokens resulting from the features fusion of local vision with global skeleton motion. Given that the evaluation of gait videos for PD can be viewed as a classification task, a Multi-Layer Perceptron (MLP) head is incorporated at the final segment of the network model. This MLP head is fundamentally composed of a sequence of fully connected layers, with the number of neurons in the output layer corresponding to the number of categories in the classification task. Specifically, if there are C categories, the final fully connected layer will have C outputs. This can be represented by the formula:

$$Score = ReLU(Output_{TF} \cdot W_1 + b_1)W_2 + b_2 \quad (8)$$

In the content provided, $Output_{TF}$ represents the output of the temporal fusion encoder and also serves as the input to the MLP Head; W_1 and W_2 denote weight matrices; b_1 and b_2 are bias terms; $Score$ signifies the evaluation results output by the final model. In our study, the scoring range for PD gait assessments is from 0 to 3, indicating that there are four categories for classification. Consequently, the final output "Score" comprises four outputs.

4. Experiments and results

4.1. Configuration and performance metrics

In this paper, we employed VIBE (Video Inference for Human Body Pose and Shape Estimation) (Kocabas et al., 2020) to extract skeleton and keypoints information from gait videos. VIBE provided reliable human joint estimations and has been widely adopted in video-based PD gait analysis studies (Endo et al., 2022; Lu et al., 2021). Compared with other human pose estimation methods, VIBE yielded more stable performance, especially in cases where patient gait was irregular or slow. This advantage was particularly important in our setting, where robustness to motion variability is critical. The link between the skeleton and the articulations in our method was derived directly from the VIBE-based skeletal structure. All experiments in this section were conducted on a server with four Nvidia GTX 4090 GPUs, within a deep learning environment based on Pytorch. The model designs utilized in these experiments incorporated the cross-entropy loss function and employed the Adam optimizer, with a training schedule set for 50 epochs. At the start of training, the learning rate was set to 0.001, with a batch size of 16. To prevent overfitting, a dropout rate of 0.3 was applied. The entire training process took about 3.5 hours to complete. For data input, a uniform sampling method was used to input 16 frames per video. The proposed model framework was pretrained on the public dataset NTU RGB+D. The choice of NTU RGB+D for pretraining was driven by the availability of the requisite skeleton annotation data and video image data within this dataset (Shahroudy et al., 2016). Within the NTU RGB+D dataset, we excluded action categories involving multiple person interactions, opting instead for categories featuring single individuals and minimal background interference.

To evaluate the performance and efficacy of our proposed methods, we presented four key metrics used in the experiment: the average F1 score, the area under the Receiver Operating Characteristic (ROC) curve (AUC), precision (Pre), and recall (Rec). We conducted an exhaustive comparative study of the experimental results for both multi-class and binary classification tasks (i.e., classification of scores as 0 versus non-0). It was particularly noteworthy that a score of 0 in the gait assessment of PD patients did not indicate they were not afflicted by PD, but rather suggested that they might exhibit a lower severity of symptoms in this assessment. Furthermore, we employed a participant-based k-fold cross-validation method for experimental design, where k was set to 5. To ensure model generalizability and prevent information leakage, data from the same video, after undergoing data augmentation to produce new videos, were not allocated to different datasets. Table 2 provides a detailed overview of the dataset distribution involved in the experiments, specifically including the data utilized during the model training and testing processes. The upper section of the table elaborates on the distribution of original video materials from participants with PD gait and their specific counts in the cross-validation and test sets.

Table 2
The data distribution for cross-validation and test sets across 241 participants in our PD gait assessment experiment.

MDS-UPDRS Score		Score-0	Score-1	Score-2	Score-3	All
Original	Cross-vaild	25	76	88	11	200
	Test	5	15	17	4	41
	Total	30	91	105	15	241
Augmented	Cross-vaild	75	228	264	33	600
	Test	15	45	51	12	123
	Total	90	273	315	45	723

4.2. PD gait score estimation results

Table 3 presents the overall scoring accuracy of the proposed method. The results in the table were obtained by calculating the accuracy of each individual score, demonstrating that the method achieved desirable accuracy in rating the data for each score. Notably, the evaluation of data with a score of 0 was the most accurate, reaching a precision of 0.93, while the assessments for scores of 1 and 2 were less accurate. Ultimately, the method achieved an overall accuracy of 0.78 in evaluating data across all scores. These findings validate the effectiveness of the introduced approach for the assessment of gait videos in PD. The confusion matrix on the left side of Fig. 7 further illustrates the evaluation of videos across various scores by the proposed method. There was a relative abundance of confusion between scores 2 and 3, which could be attributed to the similar gait performances of these two scores, making it challenging for the model to distinguish between them. Overall, the model demonstrated better performance in the evaluation of scores 0 and 3, showing a higher accuracy compared to scores 1 and 2.

Table 4 present the performance metrics of the proposed method for PD gait videos across various score (Score-0 to Score-3), including Pre, Rec, F1, and AUC values. For the Pre metric, the model exhibited the lowest precision at 0.55 for the Score-3 category, while achieving the highest precision of 0.86 for the Score-1 category. This indicated that the model is most accurate in predicting Score-1. Regarding Rec, the model had the highest recall rate of 0.93 for the Score-0 category. The lowest recall rate was observed for the Score-1 category, at 0.71. The F1 score, a harmonic means of precision and recall, reflected a balance between these metrics. The highest F1 score was observed for the Score-2 category at 0.81, while the lowest was for the Score-3 category at 0.67. Additionally, the AUC value, an indicator of overall model performance related to the ROC curve, showed that the Score-0 category had the highest AUC value of 0.95, demonstrating the model's excellent evaluative capability for this category. The lowest AUC value was for the Score-1 category at 0.78. Overall, the model performed best in the Score-0 category, exhibiting the highest recall and AUC values. Although the Score-1 category had a higher precision, its comprehensive performance metrics (recall, F1 score, and AUC values) were relatively lower. The Score-2 category demonstrated relatively balanced performance metrics, while the Score-3 category, despite poor precision, showed relatively good performance in terms of recall and AUC values. This suggested that the features characteristic of Score-3 were more pronounced and easily recognized by the proposed model.

The ROC curve graph presented on the right side of Fig. 7 demonstrates the performance of the PD gait assessment model. Four curves were observable in the graph, each representing a different category of PD gait assessment (Scores 0–3), with each color corresponding to the ROC curve of a particular assessment category. The ROC curve for Score 0 had the highest AUC value (0.95), indicating the model's near-perfect discriminative ability for this score. In contrast, the ROC curve for Score 1 had the lowest AUC (0.78), suggesting mediocre performance, while Scores 2 and 3 had AUC values of 0.81 and 0.88, respectively, reflecting good and strong discriminative power. Overall, our proposed model demonstrated AUC values above 0.75 for all four assessment categories, indicating its commendable ability to differentiate between different gait scores. Particularly for categories scored 0 and 3, the model exhibited superior performance. Fig. 8 illustrates the predicted probability distributions for each MDS-UPDRS score class (0–3), to evaluate the model's discrimination capability. Each subplot shows a boxplot of the predicted probabilities for one target score across all ground truth categories. The Kruskal–Wallis H-statistics and associated p-values are reported to assess statistical significance. From the plots, it is evident that the model assigns higher predicted probabilities to samples with matching true labels. For instance, in the subplot for Score 0, the median predicted probability is significantly higher when the true label is 0 than when it is 1, 2, or 3. This trend is consistent across all four score groups. Notably, the p-values for all Kruskal–Wallis tests are below 0.001, indicating statistically significant differences in predicted probability distributions among the score categories. These results confirm that the model can effectively distinguish between different MDS-UPDRS scores, supporting its reliability for fine-grained PD gait severity assessment.

Table 3
The accuracy results of our method in gait assessment.

MDS-UPDRS Score	Score-0	Score-1	Score-2	Score-3
Acc	0.93	0.71	0.78	0.83
Total Acc(%)	78.05			

4.3. Ablation experiments results

In our study, the proposed model combined commonly used skeleton motion features with the newly introduced keypoints vision features. To delve into the impact of these two distinct features on the assessment of PD gait, we also conducted ablation experiments to separately analyze each set of features. As shown in Table 5, the results of the ablation experiments indicated that when skeleton features are inputted alone, the skeleton-based branch network exhibited excellent performance in gait assessment with score 0. However, its performance declined with non-0 quantifications, achieving an overall accuracy of 0.67. This indicates that skeleton motion features are highly sensitive to gait performance at both score 0 and score non-0. When the model only accepted the vision features of keypoints, its performance on scores 0 slightly deteriorated compared to the use of skeleton motion features. However, it demonstrated an improvement in the quantification during non-0 scores. The overall evaluation accuracy reached 0.72. After integrating both branch models with gait features, the assessment performance was improved across all metrics. The overall assessment accuracy was enhanced to 0.78. These results suggest that skeleton motion and keypoints vision features each exhibit different performances across various scoring categories, and their integration significantly improves the model's accuracy in each score assessments. In gait analysis, skeleton movement features demonstrate a prominent performance in distinguishing between score 0 and score non-0, whereas keypoint vision features excel in the quantitative assessment of gait between score non-0. We hypothesized that this is because skeleton motion features effectively represent global movement changes, which were more pronounced when distinguishing normal from abnormal gait. In contrast, keypoint vision features could capture subtle visual differences, such as joint tremors or asymmetries, making them more suitable for distinguishing between different levels of gait impairment (scores 1 to 3). Both modality features may contain complementary information for the assessment of PD gait videos, and combining these types of information can effectively enhance the model's accuracy in evaluating PD gait videos.

The confusion matrices in Fig. 9 illustrate the model's performance when using either skeleton motion features or keypoints vision features across different scoring levels. Compared to the model that integrated both types of features, the performance of using a single feature input was diminished across all scoring levels, with a particularly noticeable decline in performance when assessing score levels 1 to 3. Further comparison of the two confusion matrices revealed that skeletal motion features outperformed purely keypoint vision feature inputs in evaluating scores of 0 and non-0 alike. However, keypoint vision features demonstrated superior performance in differentiating non-0 score levels. This comparative result unveiled the uniqueness of the two types of features in characterizing PD gait and concurrently confirmed the efficacy of feature fusion in enhancing model accuracy for PD assessment.

To explore the specific human body keypoints targeted by our proposed model in the assessment of PD gait, we further provided visualizations that demonstrate how the dual-modality framework engages with these body parts. Fig. 10 displays the visualization results of the

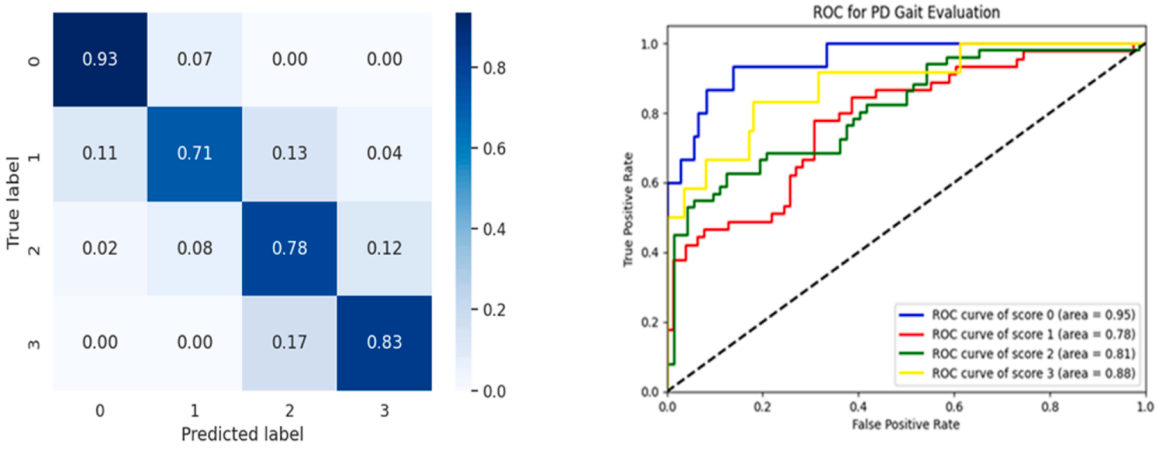


Fig. 7. Confusion matrix (left) and ROC curves (right) of PD gait assessment using our method.

Table 4
Comparison of metric results for gait assessment, including each score and the average.

MDS-UPDRS Score	Score-0	Score-1	Score-2	Score-3	Avg
Pre	0.70	0.86	0.83	0.55	0.74
Rec	0.93	0.71	0.78	0.83	0.81
F1	0.80	0.78	0.81	0.67	0.77
AUC	0.95	0.78	0.810	0.88	0.86

skeleton and keypoints attention. We obtained these visualization results from typical participants at each severity level. These visual attention maps were extracted from the final self-attention layer of the last encoder block in the Keypoints Vision Encoder. They were subsequently upsampled and visualized to highlight the focus of the model. In the skeleton visualization results, we demonstrated the visual outcomes of the skeleton sequences, highlighting the four most prominent values with circles, where larger circles represented higher values. It is observable that as the severity increases, the amplitude of skeleton motion decreases, which correlates with the rigidity of the participant's gait movements. In the keypoints visualization results, we presented the body parts focused on by the model using visual attention maps of the keypoints. The attention maps used variations in color intensity to show the impact of each region on the model's decision-making, with brighter areas indicating the model's main focus points, while darker areas received relatively less attention. From the overall results, it was evident

Table 5
Ablation experiments results for two branches in our proposed framework.

Operation	Score-0	Score-1	Score-2	Score-3	Total Acc
Skeleton motion	0.80	0.64	0.67	0.67	0.67
Keypoints vision	0.73	0.67	0.76	0.75	0.72
Dual-modality	0.93	0.71	0.78	0.83	0.78

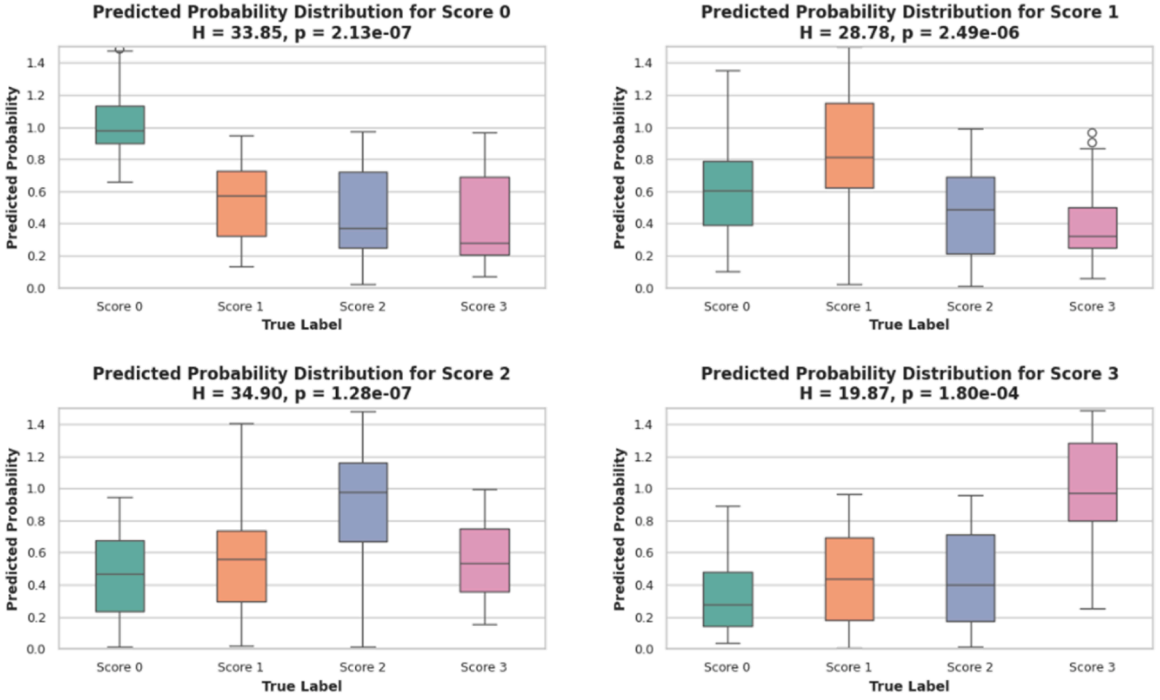
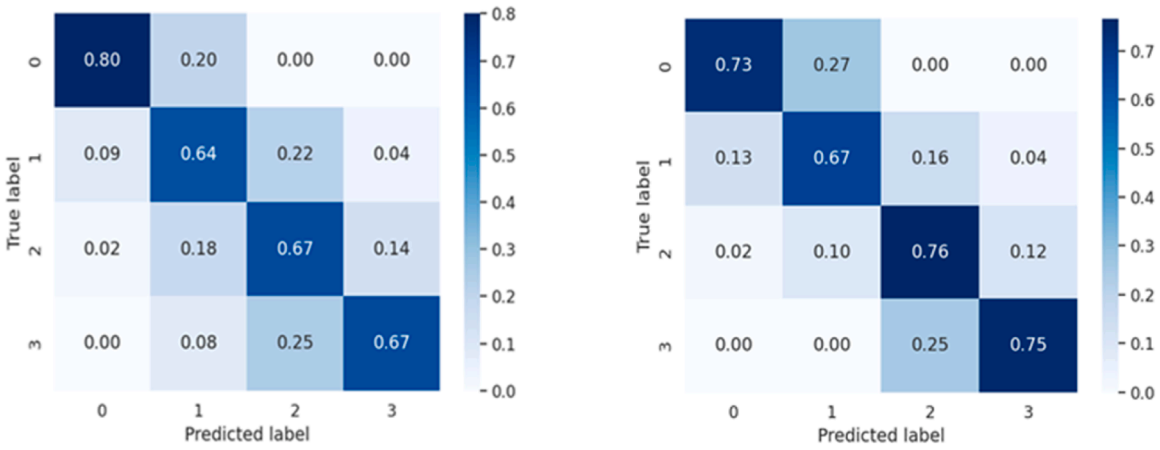


Fig. 8. Statistical analysis of predicted probabilities across every UPDRS scores. Box plots illustrating the predicted probability distributions for each MDS-UPDRS score category (Score 0–3) across all true labels. Kruskal–Wallis H-tests were conducted to evaluate the statistical significance of differences among groups. The H-statistic and associated *p*-value are reported above each plot.



(a) Confusion matrix of skeleton motion branch (b) Confusion matrix of keypoints vision branch

Fig. 9. Confusion matrices for each branch in our proposed dual-modality framework.
(a) Confusion matrix of skeleton motion branch, (b) Confusion matrix of keypoints vision branch.

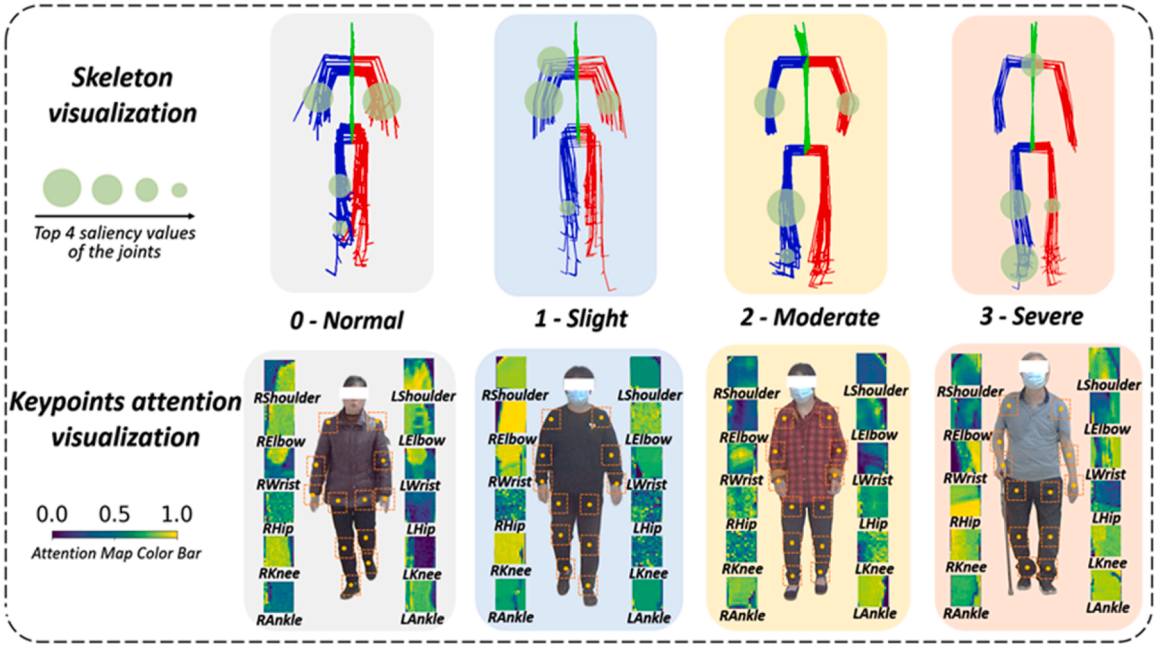


Fig. 10. Visualization results of our dual-modality framework are shown: the upper row displays skeleton feature visualizations of participants, and the lower row shows keypoints attention visualizations. These attention maps highlight the focal areas of the model, emphasizing how the model prioritizes different regions of the input image during features extraction.

that the model focuses on different body parts for participants with varying severity of symptoms. Specifically, for participants with milder symptoms, the model primarily focused on the upper limbs during gait motion. As the severity of gait impairment increased, the focus of the model shifted from the upper limbs to the lower limbs of the participants. Additionally, the dual-modality frameworks focused on different body regions for participants with varying severity of symptoms. At milder symptom stages, the regions of the body that each branch of the dual-modality frameworks focused on were relatively similar. However, as the severity increased, the areas of focus between the two branches began to diverge. Specifically, the range of body parts focused on by the keypoints vision branch expanded, which may have accounted for its superior performance in quantifying non-zero severity. These visualization results elucidated the mechanism and efficacy of our proposed framework for gait evaluation, thereby enhancing its interpretability.

4.4. The generalization results of the model

To validate the generalization capability of our proposed method, we collected gait videos recorded from more general perspectives for 26 participants. Specifically, we simulated a real-world surveillance environment by capturing gait sequences using a standard ceiling-mounted camera setup, as illustrated in the lower-left corner of Fig. 1. The camera was installed at a height of approximately 3 m, and recorded participants as they walked through a predefined area beneath the field of view. The video data was captured using the default configuration of the camera system, with a resolution of 1280×720 pixels, a frame rate of 10 frames per second, and saved in MP4 format. The genders and scores distribution of the dataset were detailed in Table 6. Each patient was recorded from two different perspectives from typical surveillance camera. Both perspectives were captured from cameras mounted at the same height, but with different horizontal and vertical angles. According

to the camera specifications, the vertical angle difference was approximately 15 degrees, and the horizontal angle difference was around 45 degrees. As a result, the total number of videos was twice the number of participating patients. It is important to note that these videos were solely utilized for testing the trained model and did not contribute to the training process. The aim of this experiment was to assess the performance of the proposed model under more general recording perspectives.

Table 7 and Fig. 11 present the results obtained by applying the trained model to the datasets in Table 6. Table 7 showcases the evaluation metrics for each score and the overall accuracy. From the table, it was evident that Score-0 maintained good performance across videos recorded from more general perspectives, indicating that our well-trained model retained high sensitivity for distinguishing between normal and abnormal gaits under more general recording perspectives. This could be attributed to the more distinct visual differences between healthy gait patterns and those with Parkinsonian symptoms, making Score-0 easier to identify even under varied recording perspectives. The performance for Score-1 and Score-2 showed a decline which suggested a reduction in quantification capability. However, the decline remained within an acceptable range. Surprisingly, there was an improvement for Score-3, which could be attributed to the smaller dataset size for this rating. Another possible explanation is that the severely impaired gait patterns in this category exhibit more prominent and recognizable features, which may have enhanced the model's prediction accuracy, even under more general recording perspectives. Ultimately, the model achieved an excellent accuracy of 73.07% on this dataset. Furthermore, the ROC curves in the left part of Fig. 11 reveal that the model performed best on data with a score of 0 and worse on data with a score of 2. The observed decline in performance can be partially attributed to occlusions of keypoints in the videos captured from certain angles. During the gait movement, vision information of some critical keypoints would be occluded by other body parts, which affected the accuracy of model predictions. In practical observation, videos recorded from a frontal view tend to experience fewer occlusions compared to those captured from other perspectives. This also explains the slight drop in accuracy observed here compared to the results reported in Section 4.2. Overall, the model demonstrated acceptable performance across all score categories.

The confusion matrix on the right side of Fig. 11 provided a detailed exposition of the model's performance in rating videos of PD gait, which were recorded from more general perspectives. Consistent with the previous two results of Table 4 and Fig. 7, the results from the confusion matrix indicated that the model exhibited high accuracy in distinguishing between score 0 and non-0. However, the model's performance in identifying between score 1 to score 3 showed a decline compared to its performance on the original dataset, though it remained within an acceptable range. These above results demonstrated that our proposed model exhibited commendable performance in assessing PD gait from more general recording perspectives. Although there was a slight decline in quantifying non-0 scores (Score 1∼3) compared to the original dataset, the model maintained high performance in distinguishing between 0 and non-0 scores. It indicated the generalization capability of our model in cross-view gait video assessment, enabling accurate evaluation of PD gait across broader recording perspective of video.

Table 6
Distribution of gait video dataset recorded from more general perspectives (26 participants).

MDS-UPDRS	Score-0	Score-1	Score-2	Score-3	All
Male/Female	1/3	4/3	5/6	2/2	14/12
Participants	4	7	11	4	26
Videos	8	14	22	8	52

Table 7
Assessment results of gait videos recorded from more general perspectives (26 participants).

MDS-UPDRS Score	Score-0	Score-1	Score-2	Score-3
Pre	1.00	0.65	0.74	0.67
Rec	0.88	0.79	0.64	0.75
F1	0.93	0.71	0.68	0.71
AUC	0.98	0.84	0.66	0.89
Total Acc (%)	73.07			

4.5. Comparison with SOTA methods

In the experimental section, a comparative analysis was conducted between our method and the existing work related to PD gait assessment. Considering the variations in evaluation criteria and outcomes across different studies, we compiled two comparative tables to showcase the differences and characteristics among various methods, juxtaposing them with the introduced approach. Table 8 presents a comparative analysis of some of the advanced PD gait assessment methods, encompassing sensor-based approaches, combinations of sensor and video techniques, and purely video-based methods. Given the limited number of participants included, most studies had adopted video data augmentation strategies. Consistent with the scoring range utilized in our research, other methods also excluded data with a score of 4. Compared to other methods listed in the table, our study featured a larger number of participants and volume of video data, achieving higher assessment accuracy within the same scoring criteria.

Table 9 presents a comparison of several advanced methods for gait assessment in PD based on video in recent years. The methods included in the table involved a relatively small number of participants and video data. The scoring ranges primarily covered binary classification (scores of 0 and non-0), 0–2, and 0–3. Unlike the evaluation metrics in Tables 4–7, the metrics shown in Table 9 (Pre, Rec, and F1) were derived by averaging the values across different score levels. From the table, it was evident that our research included the highest number of participants and video data. In binary classification assessment, the proposed method achieved the best results in the Rec metric. Although the Pre and F1 of (Gholami et al., 2023) were the highest, their method was based on a smaller cohort of participants. In a broader classification assessment, the method proposed in this study demonstrated superior performance across all evaluation metrics.

We further explored the generalization performance of our proposed method compared to current state-of-the-art methods across various gait video datasets. Due to the fact that most video-based PD gait assessment methods and their datasets were not yet open-sourced, we were only able to present comparisons of available methods that could be reproducible. We evaluated the performance of these methods on their respective original datasets and the two datasets used in our study. Table 10 showed the comparative results based on two datasets used in our study: "Ours-1″ represented the dataset of gait videos recorded from the front view, and "Ours-2" was the dataset captured from more general perspectives to simulate real-world scenarios. The results of other methods on the "Ours-1″ dataset were based on models trained on the training set and evaluated on the test set. To ensure the comparability of results, the training approaches and dataset divisions for these methods were consistent with those used in our study. The results for "Ours-2″ in the table were also derived from models tested after training on the "Ours-1″ dataset. In terms of model complexity, the method by (Lu et al., 2021) has significantly fewer parameters and lower computational cost compared to other approaches. Since our method adopts a dual-modality framework, it incurs a higher parameter count and computational cost. However, this added complexity contributes to the model's improved performance and generalizability across varying video perspectives. The evaluation results indicate that, with the exception of the method proposed by (Sabo et al., 2021), all other

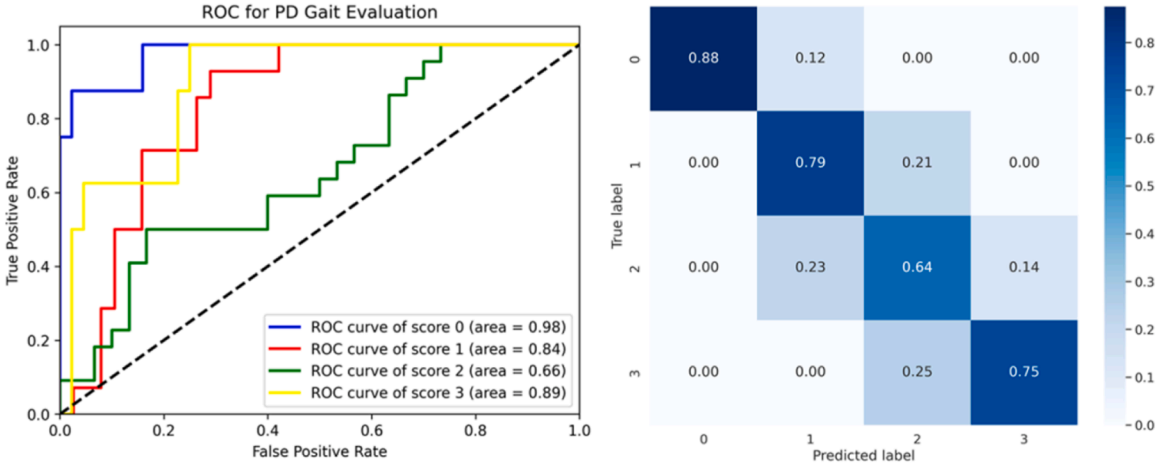


Fig. 11. ROC curves (left) and confusion matrix (right) for PD gait assessment using videos captured from generalized recording perspectives (26 participants).

Table 8
Comparison of other methods for accuracy in MDS-UPDRS gait assessment.

Methods	Data source	Participants	Video samples	Score	ACC (%)
(Parisi et al., 2015)	Sensor	34	47	(0,1,2,3)	62.0
(Parisi et al., 2016)	Sensor	38	55	(0,1,2,3)	53.0
(Sabo et al., 2020)	Sensor+Videos	14	398	(0,1,2,3)	62.1
	Sensor+Videos	14	249	(0,1,2,3)	61.4
(Guo et al., 2022a)	Videos	142	441	(0,1,2,3)	65.66
(Zeng et al., 2023)	Videos	80	480	(0,1,2,)	71.3
Ours	**Videos**	**241**	**723**	**(0,1,2,3)**	**78.05**

Table 9
Comparison of other methods for more indexes in MDS-UPDRS gait assessment.

Methods	Participants	Video samples	Score	Pre	Rec	F1
(Zeng et al., 2023)	80	480	(0,1,2)	0.713	0.710	0.710
(Guo et al., 2022a)	142	441	binary	0.726	0.720	0.723
	142	386	(0,1,2)	0.582	0.553	0.562
	142	441	(0,1,2,3)	0.654	0.651	0.649
(Lu et al., 2021)	51	–	(0,1,2)	0.667	0.683	0.661
	51	–	binary	0.68	0.72	–
(Sabo et al., 2020)	53	399	(0,1,2)	0.529	0.553	0.520
(Gholami et al., 2023)	34	–	binary	**0.90**	0.89	**0.89**
Ours	**241**	**723**	binary	0.70	**0.93**	0.80
	241	**723**	(0,1,2,3)	**0.735**	**0.813**	**0.765**

methods exhibited a decline in performance on the "Ours-1″ dataset. When tested on the "Ours-2″ dataset, all methods demonstrated a further decline in performance compared to their results on "Ours-1". This suggests that these methods possess limited generalization capabilities in cross-view gait assessment. In contrast, our proposed method exhibited a slight decline in performance, further demonstrating its superior generalization capabilities in cross-view PD gait assessment.

5. Discussion

The assessment of PD gait plays a pivotal role in the evaluation and treatment of the disease and currently represents a significant research focus within the field of PD assessment (Armstrong and Okun, 2020; Mirelman et al., 2019). With the advancement of computer vision, a variety of video-based gait assessment methods for PD have been developed. These methods primarily utilize human pose estimation to record the skeletal motion information of patients from videos, which facilitates the analysis of the gait motion process (Guo et al., 2020; Tian et al., 2024; Wu et al., 2022). However, assessing PD gait requires a meticulous approach, analysing from both global and local perspectives. A sole focus on overall skeletal dynamics risks overlooking intricate local motion details captured in the videos. To address this, we performed an in-depth analysis of the vision features of human keypoints after extracting skeletal movement information from gait videos. This method combines the analysis of skeletal motion with the vision features of human keypoints, creating a more comprehensive framework for evaluating PD gait. The improved accuracy and generalization of our method have important clinical implications. Reliable gait severity classification can assist clinicians in making more consistent and objective assessments, potentially reducing inter-rater variability associated with traditional scoring. Moreover, accurate video-based assessment under non-standard recording conditions supports flexible deployment in outpatient, community, or home environments. This is especially valuable for remote monitoring, where early detection of motor deterioration can enable more timely intervention and personalized treatment planning.

Currently, the vast majority of PD gait assessment methods rely solely on skeleton information. However, skeleton data primarily consists of coordinate information, which fails to capture vision motion of specific body keypoints during gait assessment. Such vision dynamics are crucial factors that raters focus on during gait assessments. Therefore, we first considered the manner to fully leverage the vision information in gait videos and subsequently integrated it with the skeleton motion information. In the extraction of vision information from human keypoints, utilizing the entire frame content of gait video would introduce substantial background noise and significantly increase the amount of unnecessary input data. Consequently, traditional action recognition algorithms based on skeleton and RGB images are not suitable for the assessment of PD gait videos. In fact, PD gait assessment primarily focuses on the vision information of keypoints. We only embed the patches of these human keypoints to further extract their visual features. Additionally, to further mitigate the interference from background information in the keypoint vision patches we extracted, we filtered the background from the gait videos prior to embedding the patches. This

Table 10
Classification results of our PD gait score method was compared with current open-source methods. We compared results of these methods from their papers. We also train their model on our dataset and report the results.

Methods	Dataset	Participants	Score	Parameters	Computation	Pre	Rec	F1
(Lu et al., 2021)	(Lu et al., 2021)	51	(0,1,2)	0.15 M	2.1 MFLOPs	0.67	0.68	0.66
(Sabo et al., 2021)	(Sabo et al., 2021)	53	(0,1,2)	3.08 M	16.2 GFLOPs	0.53	0.55	0.52
(Endo et al., 2022)	(Endo et al., 2022)	54	(0,1,2,3)	17.4 M	7.1 GFLOPs	0.79	0.75	0.76
(Lu et al., 2021)	Ours-1	241	(0,1,2,3)	0.15 M	2.1 MFLOPs	0.67	0.71	0.70
(Sabo et al., 2021)	Ours-1	241	(0,1,2,3)	3.08 M	16.2 GFLOPs	0.61	0.65	0.63
(Endo et al., 2022)	Ours-1	241	(0,1,2,3)	17.4 M	7.1 GFLOPs	0.60	0.62	0.62
Ours	Ours-1	241	(0,1,2,3)	18.41 M	29.1 GFLOPs	**0.74**	**0.81**	**0.77**
(Lu et al., 2021)	Ours-2	26	(0,1,2,3)	0.15 M	2.1 MFLOPs	0.57	0.55	0.55
(Sabo et al., 2021)	Ours-2	26	(0,1,2,3)	3.08 M	16.2 GFLOPs	0.54	0.55	0.54
(Endo et al., 2022)	Ours-2	26	(0,1,2,3)	17.4 M	7.1 GFLOPs	0.50	0.53	0.51
Ours	Ours-2	26	(0,1,2,3)	18.41 M	29.1 GFLOPs	**0.76**	**0.76**	**0.75**

approach not only minimizes interference from the background but also significantly reduces the volume of input data. In our analysis of model predictions in Section 4, it was observed that most misclassifications occurred between scores 1 and 2. This issue is also commonly encountered in many video-based PD motor assessments (Guo et al., 2022a; Liu et al., 2023; Lu et al., 2021). The primary reason is that these two adjacent levels represent subtle gait abnormalities that are inherently difficult to distinguish, such as minor reductions in arm swing amplitude or slight shortening of step length. In contrast, score 0 (normal gait) and score 3 (severe impairment) typically exhibit more pronounced and distinguishable features. This finding highlights that accurately differentiating the intermediate levels of PD severity remains a highly challenging task, which directly impacts the performance of fine-grained multi-score classification.

To explore whether adding these structural components would introduce redundancy, we conducted ablation studies, as shown in Table 5. The results demonstrate that the fusion of the two branches yields clear performance gains, indicating that the two modalities provide complementary rather than redundant information. Furthermore, we have investigated the two branches of our proposed framework to elucidate the mechanisms during PD gait assessment, with the objective of enhancing the interpretability of our model. Even though both branches receive coordinate and connection information, the learned attention maps reveal that they focus on different keypoint regions. This demonstrates that structural cues are interpreted differently by each branch, further supporting the notion that incorporating both adjacency and coordinate information in each branch is not redundant but mutually beneficial. Through visualization of gait features across two modalities, we observed that the body regions focused on by the model differ at varying severity levels of the condition. This indicates that different severity levels manifest distinct characteristics. Additionally, by comparing the two branches of the overall framework, it also could be discerned that the differences in gait feature representation between skeleton motion and keypoints vision. At milder symptom stages, the regions of the body focused on by each branch of the dual-modality framework are relatively similar. However, as the severity increases, the areas of focus between the two branches begin to diverge. Specifically, the range of body parts focused on by the keypoints vision branch has expanded to encompass more regions, which may account for its superior performance in quantifying non-zero severity compared to the skeletal motion branch. The analytical results reveal the performance discrepancies among body parts in the assessment of PD gait. These findings also align closely with the aspects that professional raters prioritize in PD gait assessments, which provides a scientific foundation for further understanding the mechanisms underlying PD gait assessment.

Our method showcases exceptional generalization in the cross-view analysis of PD gait videos. As shown in Table 10, a comparative analysis of performance across different datasets revealed a significant decline in the efficacy of other methods when evaluating PD gait videos from more general perspectives. In contrast, our proposed method exhibits only a slight decrease which indicates its ideal generalization. This robustness is largely due to the incorporation of vision information from human keypoints. Specifically, this includes not just the visual data but also the spatial positioning and connections of the keypoints. Furthermore, the ablation experimental results in the experimental section indicated that keypoint vision features significantly outperformed skeleton motion features when used independently for PD gait assessment. Indeed, in our proposed Keypoint Vision Transformer structure, not only is rich vision information encapsulated, but spatial location information of keypoints and their interconnections are also introduced through adjacency embeddings. These comprehensive details inherently contain the global information of the skeleton, thus providing a more holistic feature description for gait assessment. By combining skeleton motion and keypoint vision features, our model achieves significantly improved assessment accuracy.

Although our proposed method achieves promising accuracy and generalization performance, there remain several practical limitations and implementation challenges could be addressed in future work. We investigated several prevalent feature fusion approaches for dual-modality features and ultimately adopted the concatenation method due to its simplicity and superior performance. Future research endeavours could focus on the design of more efficient fusion modules to enhance the synergistic performance of integrating these two distinct feature types. Due to concerns about model parameter count, we did not adopt current complex skeleton-based feature extraction models into our branch of skeleton extraction. Instead, we chose the mainstream ST-GCN as our foundational model, which is a commonly used and effective baseline in current research (Sabo et al., 2022, 2021). In future work, consideration could be given to using more streamlined and efficient models for skeleton information extraction as a potential substitute. Additionally, as our proposed framework incorporates vision information, the vision information of keypoints in gait videos are inevitably occluded, which could affect the final performance. Future work could consider employing generative networks to reconstruct the occluded vision information based on temporal sequences of frames. While offering a generalizable and interpretable framework for PD gait assessment, the dual-branch structure and incorporation of video sequences inevitably increase model complexity and resource requirements. These factors pose significant challenges for deployment in real-world clinical applications, particularly on resource-limited clinical devices or edge computing platforms. Future work will focus on simplified model architectures and efficient feature fusion to reduce computational complexity and facilitate practical clinical implementation. Overall, compared with existing methods, our proposed method exhibited significant advantages in video-based assessments of PD under more general conditions. It also offers new perspectives and methodologies for further advancing remote monitoring and assessment of PD in home and community settings.

6. Conclusion

In this study, we propose a novel dual-modality framework based on human keypoints vision and skeleton motion information for the assessment of PD gait. This framework extracts PD gait features in videos from two perspectives. It extracts the vision features of human body keypoints through the proposed Transformer-based model, which also encompasses the positional and connectivity information of keypoints. After extracting the skeleton motion information, we introduce a temporal fusion encoder to integrate these two types of motion features and extract the temporal dynamic characteristics of the gait. The final experimental results indicate that our proposed model demonstrates superior advancement over existing methods for the assessment of PD gait videos. To demonstrate the interpretability of our method, we visualize the features of the two framework branches, which also revealed the mechanisms of human keypoints in the PD gait assessment. We additionally apply the trained model to gait videos recorded from more general perspectives and observe that the model continues to demonstrate satisfactory performance. This outcome further substantiates its exceptional generalization capability in the assessment of cross-view videos. Despite its promising results, our method has several limitations. The dual-branch structure increases the model's computational complexity, which may hinder deployment in real-time or resource-constrained clinical environments. In addition, keypoint occlusion in non-frontal views may affect visual feature extraction, potentially impacting assessment accuracy. In future work, we aim to explore lightweight network architectures, improved feature fusion mechanisms, and generative techniques for recovering occluded keypoint features, in order to reduce complexity and facilitate practical clinical implementation.

CRediT authorship contribution statement

Weiping Liu: Writing – original draft, Validation, Writing – review & editing, Visualization, Software, Formal analysis, Methodology, Data curation. **Xiaozhen Lin:** Data curation, Formal analysis. **Xinghong Chen:** Data curation. **Yifang Liu:** Data curation. **Zengxin Zhong:** Data curation. **Rong Chen:** Supervision. **Guannan Chen:** Validation, Resources, Methodology, Formal analysis, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Yu Lin:** Writing – review & editing, Resources, Funding acquisition, Conceptualization, Supervision, Project administration, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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